

AI Usage Note

This document summarizes how Artificial Intelligence tools were utilized during the development of the Flaky Test Detector & Explainer project, in accordance with the hackathon submission guidelines.

AI Tools Used

- ChatGPT
- Claude
- Gemini
- Ollama (Local Llama 3)

1. What AI Helped With

AI was actively utilized throughout the software development lifecycle to accelerate implementation and enhance the final product. Key areas of assistance included:

- Project Architecture Design
- Streamlit Dashboard Development
- Agent Workflow Implementation
- Database Integration
- Unit Test Generation
- Universal Parser Engine
- API Fallbacks
- Documentation
- Prompt Engineering

2. What AI Got Wrong

- Dependency and Syntax Hallucinations
- Simplistic Test Assertions
- LLM Output Formatting Issues

3. Best Prompts Used

For System Architecture & Boilerplate Generation:

"You are a Senior Python Software Engineer. I am building a hackathon project called 'Flaky Test Detector & Explainer'. The goal is to analyze test execution history, detect flaky tests, and use a local LLM to explain root causes. Please generate the complete project folder structure and architecture overview using best practices."

For Comprehensive Pytest Generation:

"Write a comprehensive pytest test suite for the detector.py module. Include fixtures for temporary CSV files and in-memory pandas DataFrames. Ensure you test successful execution, missing files, empty files, invalid schemas, statistics calculation, flakiness detection logic, and severity classification. Target a 100% pass rate."

For LLM Root-Cause Analysis (System Prompt):

"You are a senior QA engineer. Analyze the following flaky test. Test Name: {test_name} Flakiness Score: {score} Failure Logs: {logs} Provide: 1. Likely Root Cause 2. Confidence Percentage 3. Recommended Fix"